

International Meeting Chile–Honduras

Differential Equations in Biological Sciences and Engineering

August 7, 2026

Virtual Event

Program

Morning Session

Time	Activity
09:00 – 09:10	Welcome Remarks Dr. Samuel Ortega-Farías (CITRA–UTALCA, Chile)
	<i>Morning Session Moderator: Dr. Marlon M. López-Flores</i>
09:10 – 09:40	Plenary Lecture 1 <i>Asymptotic Behavior of a Logistic Epidemiological Model with Random Transmission and Treatment Rates</i> Dr. María Clara Fittipaldi (UNAM, México)
09:50 – 10:20	Plenary Lecture 2 <i>Limit Cycles on Hybrid Systems</i> Dr. Ángela Carolina Tunubalá Sánchez (UFG, Brazil)
10:30 – 10:45	Coffee Break
10:45 – 11:15	Plenary Lecture 3 <i>What Is the Smoothest Local Average?</i> Dr. José Ramón Madrid Padilla (Virginia Tech, USA)
11:25 – 11:55	Plenary Lecture 4 <i>A GAMLSS-Based Approach for Modeling Fruit Weight Distributions in ‘Hayward’ Kiwifruit Orchards</i> Dr. Emilio Vilches Gutiérrez (UOH, Chile)
12:05 – 12:20	<i>Highlighting the Bonds of Solidarity Between Honduras and Chile</i> Dra. Rosibel Pacheco Banegas (ASOMATH, Honduras)
12:20 – 12:30	General Discussion

Afternoon Session

Time	Activity
<i>Afternoon Session – Block I Moderator: Dr. William Campillay-Llanos</i>	
14:30 – 14:40	<i>A Proportional Positive-State Model for Post-Irrigation Soil-Moisture Recovery Using Open Smart-Irrigation Data</i> Lic. Gustavo Maldonado (UNAH, Honduras)
14:45 – 14:55	<i>Multiyear Validation and Mass-Balance Auditing of Richards Simulations on Uniform and Graded Finite-Volume Meshes Using USDA SCAN Observations</i> Lic. José David Zúñiga (UNAH, Honduras)
15:00 – 15:10	<i>Proportional Discrete Fourier Analysis of WaPOR Actual Evapotranspiration over Cropland in Comayagua, Honduras</i> Lic. Carlos Miguel Cruz (UNAH, Honduras)
15:15 – 15:25	<i>A Nonlocal Spatio-Temporal Framework for Conidial Dispersal and Infection Risk in Apple Orchards</i> Dr. Marlon M. López-Flores (LIARC-IME, Brazil)
15:30 – 15:45	Coffee Break
<i>Afternoon Session – Block II Moderator: Dr. William Campillay-Llanos</i>	
15:45 – 15:55	<i>From Classical Phenological Models to Reaction-Diffusion Equations: A Spatial Bioclimatic Framework for Cherry Phenology</i> Dr. William Campillay-Llanos (NIPA-UCT, Chile)
16:00 – 16:10	<i>More than 25 Years of Biomathematical Modelling: Advances and Perspectives from the CITRA Experience</i> Dr. Samuel Ortega-Farías (CITRA-UTALCA, Chile)
16:15 – 16:25	<i>Background and Experiences in the Use of Biomathematical Tools in Irrigation Science in Southern Chile</i> Dr. Rafael López-Olivari (INIA, Chile)
<i>Afternoon Session – Block III Moderator: Dr. Marlon López-Flores</i>	
16:30 – 16:40	<i>Origin and Maintenance of Biological Diversity</i> Dr. Géza Meszéna (ELTE & HUN-REN CER, Hungary)
16:45 – 16:55	<i>Presentation of the JONRAF Special Issue</i> Dr. Braulio Soto (UCT, Chile)
16:55 – 17:00	Closing Remarks

Organized by: Faculty of Agricultural Sciences and CITRA, Universidad de Talca
With the participation and support of: NIPA-UCT, JONRAF, SOMACHI and ASOMATH